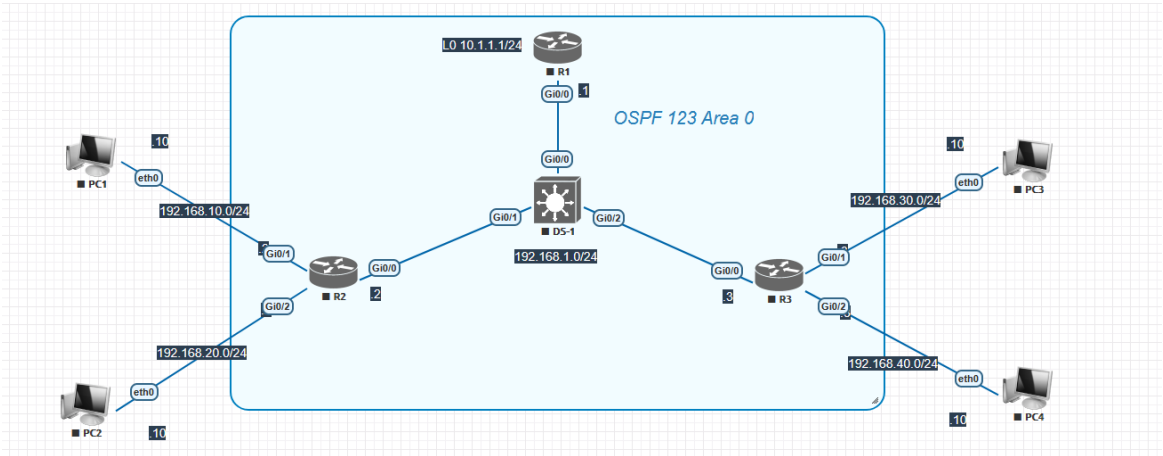


OSPF Lab Manual

Configure & Verify OSPF Process 123 — Area 0

Topology: R1, R2, R3, DS-1 • 4 PCs • Single Area



Lab Topic	OSPFv2 Single-Area (Area 0)
OSPF Process ID	123
Area	Area 0 (Backbone)
Devices	R1, R2, R3, DS-1 (L3 Switch), PC1–PC4
Difficulty	Intermediate

1. Lab Objectives

Upon completing this lab, you will be able to:

- Configure OSPFv2 process 123 on routers R1, R2, R3, and Layer 3 switch DS-1
- Advertise all directly connected networks into OSPF Area 0
- Configure and verify OSPF router IDs using loopback interfaces
- Verify OSPF neighbor adjacencies and database synchronization
- Confirm end-to-end connectivity between all PCs across the OSPF domain
- Use show and debug commands to troubleshoot OSPF operations
- Understand DR/BDR election on multi-access segments

2. Topology Overview

The topology consists of four OSPF-capable devices inside a single backbone area (Area 0) running OSPF process 123. Four PCs connect to the edge routers as access networks that must be reachable across the OSPF domain.

2.1 Network Diagram Description

DS-1 is a Layer 3 switch acting as the central hub. R1, R2, and R3 each connect into DS-1 on separate physical ports (Gi0/0, Gi0/1, Gi0/2 respectively), but all three ports are switched together as access ports in VLAN 10. DS-1's Vlan10 SVI (192.168.1.254/24) is the only Layer 3 address on the switch for this segment, making 192.168.1.0/24 a single shared multi-access broadcast domain for OSPF.

2.2 IP Address Table

Device	Interface	IP Address	Subnet	Connected To
R1	Gi0/0	192.168.1.1	/24	DS-1 Vlan10 (via Gi0/0)
R1	Loopback0	10.1.1.1	/24	Router ID
R2	Gi0/0	192.168.1.2	/24	DS-1 Vlan10 (via Gi0/1)
R2	Gi0/1	192.168.10.x	/24	PC1 LAN
R2	Gi0/2	192.168.20.x	/24	PC2 LAN
R3	Gi0/0	192.168.1.3	/24	DS-1 Vlan10 (via Gi0/2)
R3	Gi0/1	192.168.30.x	/24	PC3 LAN
R3	Gi0/2	192.168.40.x	/24	PC4 LAN
DS-1	Gi0/0	(no IP - access port, VLAN 10)	--	R1

DS-1	Gi0/1	(no IP - access port, VLAN 10)	--	R2
DS-1	Gi0/2	(no IP - access port, VLAN 10)	--	R3
DS-1	Vlan10 (SVI)	192.168.1.254	/24	OSPF backbone segment
PC1	eth0	192.168.10.10	/24	R2 Gi0/1
PC2	eth0	192.168.20.10	/24	R2 Gi0/2
PC3	eth0	192.168.30.10	/24	R3 Gi0/1
PC4	eth0	192.168.40.10	/24	R3 Gi0/2

IP Address Note

DS-1 connects to R1, R2, and R3 over a single Layer 2 VLAN (VLAN 10). Its physical ports (Gi0/0-Gi0/2) act as access ports with no IP address; the Vlan10 SVI holds the single IP (192.168.1.254/24) that participates in OSPF on the shared 192.168.1.0/24 segment. R1, R2, and R3 each connect their Gi0/0 directly to this segment as 192.168.1.1, .2, and .3.

2.3 OSPF Role Summary

Device	OSPF Role	Router ID	Networks Advertised
R1	ABR / Internal	10.1.1.1	192.168.1.0/24, 10.1.1.0/24
R2	Internal Router	Highest active IP	192.168.1.0/24, 192.168.10.0/24, 192.168.20.0/24
R3	Internal Router	Highest active IP	192.168.1.0/24, 192.168.30.0/24, 192.168.40.0/24
DS-1	Internal Router	Highest active IP (Vlan10)	192.168.1.0/24

3. Pre-Lab Requirements

3.1 Prerequisites

- Cisco IOS knowledge: interface configuration, basic routing concepts
- Ability to cable devices according to the topology diagram
- Console access to all four network devices
- Packet Tracer 8.x or GNS3 / physical lab equipment

3.2 Required Equipment

Item	Quantity	Notes
Cisco Router (e.g. ISR 4000 or PT equivalent)	3	R1, R2, R3
Cisco Layer 3 Switch (e.g. 3560/3650 or PT equivalent)	1	DS-1
PCs or VPCs	4	PC1–PC4
Ethernet patch cables	8+	Per topology links
Console cables + terminal app	4	One per device

3.3 OSPF Concept Review

Before starting, ensure you understand the following OSPF fundamentals:

OSPF Key Concepts

OSPF (Open Shortest Path First) is a link-state routing protocol. Routers flood Link State Advertisements (LSAs) to build an identical LSDB. Each router independently runs Dijkstra's SPF algorithm to compute loop-free best paths. OSPF forms adjacencies via Hello packets and uses cost (based on bandwidth) as its metric.

OSPF Term	Description
Process ID	Locally significant ID (1–65535). Neighbors can have different Process IDs.
Router ID (RID)	32-bit identifier. Priority: manual > loopback IP > highest active interface IP
Area 0	Backbone area. All OSPF areas must connect to Area 0.
Hello Interval	Default 10s on broadcast, 30s on NBMA. Must match between neighbors.
Dead Interval	Default 4x Hello interval. Neighbor declared dead if no Hello received.
DR / BDR	Designated / Backup Designated Router elected on multi-access segments.

LSA	Link State Advertisement — describes router links and their costs.
LSDB	Link State Database — must be identical across all routers in an area.
SPF	Shortest Path First (Dijkstra) algorithm — computes best paths from LSDB.

4. Device Configuration

Follow the steps below in order. Complete interface IP addressing before enabling OSPF. All OSPF configuration is under OSPF process 123.

4.1 R1 Configuration

Step 1: Configure Hostname and Interfaces

```
R1> enable
R1# configure terminal
R1(config)# hostname R1

! Configure Loopback for Router ID
R1(config)# interface Loopback0
R1(config-if)# ip address 10.1.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit

! Configure uplink to DS-1
R1(config)# interface GigabitEthernet0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# description Link-to-DS-1-VLAN10
R1(config-if)# exit
```

Step 2: Enable OSPF on R1

```
R1(config)# router ospf 123
R1(config-router)# router-id 10.1.1.1

! Advertise backbone link
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0

! Advertise loopback (for reachability to Router ID)
R1(config-router)# network 10.1.1.0 0.0.0.255 area 0

R1(config-router)# end
R1# write memory
```



Wildcard Mask Reminder

The network statement uses a wildcard mask (inverse of subnet mask). For /24 (255.255.255.0) use 0.0.0.255. For /32 (a single host) use 0.0.0.0. The network command matches all interfaces whose IP falls within the stated range.

4.2 R2 Configuration

Step 1: Configure Hostname and Interfaces

```
R2> enable
R2# configure terminal
R2(config)# hostname R2

! Backbone interface toward DS-1
R2(config)# interface GigabitEthernet0/0
R2(config-if)# ip address 192.168.1.2 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# description Link-to-DS-1-VLAN10
R2(config-if)# exit

! PC1 LAN
R2(config)# interface GigabitEthernet0/1
R2(config-if)# ip address 192.168.10.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# description PC1-LAN
R2(config-if)# exit

! PC2 LAN
R2(config)# interface GigabitEthernet0/2
R2(config-if)# ip address 192.168.20.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# description PC2-LAN
R2(config-if)# exit
```

Step 2: Enable OSPF on R2

```
R2(config)# router ospf 123

! Advertise all three networks
R2(config-router)# network 192.168.1.0 0.0.0.255 area 0
R2(config-router)# network 192.168.10.0 0.0.0.255 area 0
R2(config-router)# network 192.168.20.0 0.0.0.255 area 0

R2(config-router)# end
R2# write memory
```

4.3 R3 Configuration

Step 1: Configure Hostname and Interfaces

```
R3> enable
R3# configure terminal
R3(config)# hostname R3

! Backbone interface toward DS-1
R3(config)# interface GigabitEthernet0/0
R3(config-if)# ip address 192.168.1.3 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# description Link-to-DS-1-VLAN10
R3(config-if)# exit
```

```
! PC3 LAN
R3(config)# interface GigabitEthernet0/1
R3(config-if)# ip address 192.168.30.1 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# description PC3-LAN
R3(config-if)# exit

! PC4 LAN
R3(config)# interface GigabitEthernet0/2
R3(config-if)# ip address 192.168.40.1 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# description PC4-LAN
R3(config-if)# exit
```

Step 2: Enable OSPF on R3

```
R3(config)# router ospf 123

R3(config-router)# network 192.168.1.0 0.0.0.255 area 0
R3(config-router)# network 192.168.30.0 0.0.0.255 area 0
R3(config-router)# network 192.168.40.0 0.0.0.255 area 0

R3(config-router)# end
R3# write memory
```

4.4 DS-1 (Layer 3 Switch) Configuration

Why SVI Instead of Routed Ports?

DS-1 acts as the central hub for the 192.168.1.0/24 backbone shared by R1, R2, and R3. If you assign Gi0/0, Gi0/1, and Gi0/2 as separate routed interfaces (no switchport), IOS rejects the 2nd and 3rd assignments with an overlapping address error because all three are in the same /24. The correct approach is to keep the three ports as Layer 2 access ports in a VLAN, then assign the single IP to the VLAN SVI (interface Vlan10). DS-1 participates in the shared segment via that one logical IP.

Step 1: Enable IP Routing and Create the Backbone VLAN

```
DS-1> enable
DS-1# configure terminal
DS-1(config)# hostname DS-1

! Enable Layer 3 routing (required on all L3 switches)
DS-1(config)# ip routing

! Create the backbone VLAN
DS-1(config)# vlan 10
DS-1(config-vlan)# name OSPF-BACKBONE
DS-1(config-vlan)# exit
```

Step 2: Configure Physical Ports as Access Ports in VLAN 10

```
! Port toward R1
DS-1(config)# interface GigabitEthernet0/0
DS-1(config-if)# switchport mode access
DS-1(config-if)# switchport access vlan 10
```

```
DS-1(config-if)# no shutdown
DS-1(config-if)# description Link-to-R1
DS-1(config-if)# exit

! Port toward R2
DS-1(config)# interface GigabitEthernet0/1
DS-1(config-if)# switchport mode access
DS-1(config-if)# switchport access vlan 10
DS-1(config-if)# no shutdown
DS-1(config-if)# description Link-to-R2
DS-1(config-if)# exit

! Port toward R3
DS-1(config)# interface GigabitEthernet0/2
DS-1(config-if)# switchport mode access
DS-1(config-if)# switchport access vlan 10
DS-1(config-if)# no shutdown
DS-1(config-if)# description Link-to-R3
DS-1(config-if)# exit
```

Step 3: Assign IP Address to the SVI

```
! SVI for VLAN 10 -- DS-1's single IP on the 192.168.1.0/24 backbone
DS-1(config)# interface Vlan10
DS-1(config-if)# ip address 192.168.1.254 255.255.255.0
DS-1(config-if)# no shutdown
DS-1(config-if)# description OSPF-Backbone-SVI
DS-1(config-if)# exit

! Verify SVI is up (requires at least one VLAN 10 access port to be up)
DS-1# show interfaces Vlan10
DS-1# show vlan brief
```

Step 4: Enable OSPF on DS-1

```
DS-1(config)# router ospf 123

! Advertise backbone segment via the SVI
DS-1(config-router)# network 192.168.1.0 0.0.0.255 area 0

DS-1(config-router)# end
DS-1# write memory
```

☒ Backbone IP Summary After Fix

All four devices share 192.168.1.0/24 via VLAN 10 on DS-1: R1 Gi0/0 = 192.168.1.1/24, R2 Gi0/0 = 192.168.1.2/24, R3 Gi0/0 = 192.168.1.3/24, DS-1 Vlan10 = 192.168.1.254/24. No overlap errors because DS-1 presents a single SVI on the segment, not three separate routed ports.

4.5 PC Default Gateway Configuration

Configure each PC with the correct default gateway so traffic can route across the OSPF domain:

PC	IP Address	Subnet Mask	Default Gateway
----	------------	-------------	-----------------

PC1	192.168.10.10	255.255.255.0	192.168.10.1 (R2 Gi0/1)
PC2	192.168.20.10	255.255.255.0	192.168.20.1 (R2 Gi0/2)
PC3	192.168.30.10	255.255.255.0	192.168.30.1 (R3 Gi0/1)
PC4	192.168.40.10	255.255.255.0	192.168.40.1 (R3 Gi0/2)

5. Verification & Testing

Use the following commands to systematically verify OSPF operation. Work through them in order — each step confirms a different layer of OSPF functionality.

5.1 Verify OSPF Neighbors

Run on each router. All OSPF-enabled devices should appear in the neighbor table with state FULL.

```
R1# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.1.2	1	FULL/DR	00:00:35	192.168.1.2	Gi0/0
192.168.1.3	1	FULL/BDR	00:00:37	192.168.1.3	Gi0/0
192.168.1.254	1	FULL/DROTHER	00:00:38	192.168.1.254	Gi0/0

☒ Expected Neighbor States

On the 192.168.1.0/24 broadcast segment, OSPF will elect a DR and BDR. All routers should reach FULL state with the DR and BDR. DROTHER routers show 2WAY with each other (which is normal and correct). If any neighbor is stuck in INIT, EXSTART, or EXCHANGE, proceed to Section 7 (Troubleshooting).

5.2 Verify OSPF Interfaces

```
R1# show ip ospf interface brief
```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs F/C
Lo0	123	0	10.1.1.1/24	1	LOOP	0/0
Gi0/0	123	0	192.168.1.1/24	1	BDR	3/3

Verify the interface detail for Hello/Dead interval, network type, and DR/BDR:

```
R1# show ip ospf interface GigabitEthernet0/0
```

```
GigabitEthernet0/0 is up, line protocol is up
  Internet Address 192.168.1.1/24, Area 0, Attached via Network Statement
  Process ID 123, Router ID 10.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 192.168.1.2, Interface address 192.168.1.2
  Backup Designated Router (ID) 10.1.1.1, Interface address 192.168.1.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:05
  Supports Link-local Signaling (LLS)
  Neighbor Count is 3, Adjacent neighbor count is 3
```

5.3 Verify the OSPF Link State Database (LSDB)

The LSDB must be identical on all routers within Area 0. Run on any router:

```

R1# show ip ospf database

        OSPF Router with ID (10.1.1.1) (Process ID 123)

        Router Link States (Area 0)

Link ID      ADV Router    Age  Seq#       Checksum Link count
10.1.1.1     10.1.1.1      142  0x80000003 0x00XXXX 2
192.168.1.2  192.168.1.2   138  0x80000004 0x00XXXX 3
192.168.1.3  192.168.1.3   135  0x80000003 0x00XXXX 3
192.168.1.254 192.168.1.254 140  0x80000002 0x00XXXX 1

        Net Link States (Area 0)

Link ID      ADV Router    Age  Seq#       Checksum
192.168.1.2  192.168.1.2   138  0x80000001 0x00XXXX

```

i LSDB Verification

You should see one Router LSA (Type 1) per OSPF router in the area, and one Network LSA (Type 2) originated by the DR for the broadcast segment. Verify the same LSAs appear on all routers. Use 'show ip ospf database router <router-id>' to inspect individual LSAs.

5.4 Verify the Routing Table

OSPF-learned routes appear with code 'O'. All OSPF networks should be visible on every router:

```

R1# show ip route ospf

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

192.168.10.0/24 [110/2] via 192.168.1.2, 00:05:23, GigabitEthernet0/0
192.168.20.0/24 [110/2] via 192.168.1.2, 00:05:23, GigabitEthernet0/0
192.168.30.0/24 [110/2] via 192.168.1.3, 00:04:51, GigabitEthernet0/0
192.168.40.0/24 [110/2] via 192.168.1.3, 00:04:51, GigabitEthernet0/0

```

Prefix	Expected on R1?	Expected on R2?	Expected on R3?
192.168.1.0/24	C (connected)	C (connected)	C (connected)
10.1.1.0/24	C (connected)	O (via OSPF)	O (via OSPF)
192.168.10.0/24	O (via OSPF)	C (connected)	O (via OSPF)
192.168.20.0/24	O (via OSPF)	C (connected)	O (via OSPF)
192.168.30.0/24	O (via OSPF)	O (via OSPF)	C (connected)
192.168.40.0/24	O (via OSPF)	O (via OSPF)	C (connected)

5.5 End-to-End Connectivity Tests

Ping between all PC subnets to confirm full reachability. Run from each PC:

```
PC1> ping 192.168.20.10    ! PC1 to PC2
PC1> ping 192.168.30.10    ! PC1 to PC3
PC1> ping 192.168.40.10    ! PC1 to PC4

PC3> ping 192.168.10.10    ! PC3 to PC1
PC3> ping 192.168.20.10    ! PC3 to PC2
PC3> ping 192.168.40.10    ! PC3 to PC4
```

 Success Criteria

All pings should succeed (0% packet loss). If pings fail, verify: (1) OSPF routes exist on both source and destination routers, (2) PC default gateways are correctly configured, (3) no access-lists are blocking traffic.

5.6 Additional Verification Commands

Command	Purpose
show ip protocols	Verify OSPF process, router ID, networks, and timers
show ip ospf	OSPF process details, SPF run count, area info
show ip ospf neighbor detail	Full neighbor detail including Dead timer
show ip ospf database router	View specific Router LSA content
show ip ospf statistics	SPF calculation frequency and event log
debug ip ospf events	Real-time OSPF event logging (use with caution)
debug ip ospf adj	Adjacency formation debug (use in lab only)
show ip ospf border-routers	Show intra-area and inter-area routes to ABRs/ASBRs

5.7 Verify the DS-1 SVI and VLAN Configuration

Since DS-1 reaches the OSPF backbone through the Vlan10 SVI, confirm the VLAN, access ports, and SVI status before troubleshooting OSPF on DS-1:

```
DS-1# show vlan brief

VLAN  Name                Status    Ports
----  -
1     default                active
10    OSPF-BACKBONE          active    Gi0/0, Gi0/1, Gi0/2

DS-1# show interfaces Vlan10
Vlan10 is up, line protocol is up
  Internet address is 192.168.1.254/24

DS-1# show ip interface brief | include Vlan10
Vlan10          192.168.1.254  YES manual up
```

☒ **SVI Up/Down Behavior**

The Vlan10 SVI only comes up if at least one access port assigned to VLAN 10 is in an up/up state. If Vlan10 shows down/down, check that Gi0/0, Gi0/1, or Gi0/2 are not shutdown and are correctly assigned to VLAN 10 using show vlan brief.

6. Optional Enhancements

Once basic OSPF is working, apply these enhancements to deepen your understanding:

6.1 Configure OSPF Priorities (DR/BDR Election)

By default, the router with the highest Router ID wins the DR election. You can influence this with the `ip ospf priority` command. Set R1 as the preferred DR on the backbone segment:

```
R1(config)# interface GigabitEthernet0/0
R1(config-if)# ip ospf priority 100

! Set DS-1 as BDR candidate (configure on the Vlan10 SVI)
DS-1(config)# interface Vlan10
DS-1(config-if)# ip ospf priority 50

! Trigger re-election by clearing the OSPF process
R1# clear ip ospf process
! Answer 'yes' to the prompt
```

DR/BDR Re-election

OSPF does NOT pre-empt an existing DR/BDR. Changing the priority takes effect only after the current DR or BDR goes down, or after you clear the OSPF process. Clearing the OSPF process drops all adjacencies briefly — do not do this in production.

6.2 Configure OSPF Cost

Influence path selection by manually setting interface cost. Lower cost is preferred:

```
! Set cost on R2's backbone interface
R2(config)# interface GigabitEthernet0/0
R2(config-if)# ip ospf cost 10

! Verify cost change in routing table
R2# show ip route ospf
R2# show ip ospf interface GigabitEthernet0/0
```

6.3 Configure Passive Interfaces

PC-facing interfaces do not need to send OSPF Hellos. Mark them passive to suppress Hello packets while still advertising the network:

```
R2(config)# router ospf 123
R2(config-router)# passive-interface GigabitEthernet0/1
R2(config-router)# passive-interface GigabitEthernet0/2

R3(config)# router ospf 123
R3(config-router)# passive-interface GigabitEthernet0/1
R3(config-router)# passive-interface GigabitEthernet0/2
```

i Best Practice

Always configure passive-interface on stub LAN interfaces (connected to end hosts). This prevents unnecessary OSPF Hello traffic on host segments and eliminates the risk of a rogue device forming an OSPF adjacency.

6.4 Advertise Default Route

If R1 connects to an upstream ISP, redistribute a default route into OSPF:

```
R1(config)# ip route 0.0.0.0 0.0.0.0 <upstream-gateway>
R1(config)# router ospf 123
R1(config-router)# default-information originate

! Verify default route on other routers
R2# show ip route | include O*E
```

7. Troubleshooting Guide

Use this section when OSPF neighbors do not form or routes are missing.

7.1 Neighbor Stuck in INIT State

Symptom

show ip ospf neighbor shows a neighbor in INIT state. This means Hellos are being received but the local router's own Router ID is not in the Hello from the neighbor (unidirectional communication).

Check	Command	Fix
Physical/L1	show interfaces Gi0/0	Verify line protocol is up
IP connectivity	ping <neighbor-ip>	Fix IP addressing or cable
Same subnet?	show ip interface brief	Ensure both ends are /24 on same network
Same Area?	show ip ospf interface	Area must match (both Area 0)
Same Hello timer?	show ip ospf interface	Default 10s — must match
Same MTU?	show interfaces	Mismatch causes EXSTART stuck state

7.2 Neighbor Stuck in EXSTART/EXCHANGE

Typically caused by an MTU mismatch. One side sends DBD packets that are too large for the other:

```
! Check MTU on both sides
R1# show interfaces GigabitEthernet0/0 | include MTU

! Quick workaround: ignore MTU in OSPF (lab use only)
R1(config)# interface GigabitEthernet0/0
R1(config-if)# ip ospf mtu-ignore
```

7.3 Routes Missing from Routing Table

Symptom	Likely Cause	Verification Command
Network not in OSPF LSDB	network statement missing or wrong wildcard	show ip ospf database
Route in LSDB but not routing table	Summarization or filtering	show ip route ospf; show ip prefix-list
Route present but unreachable	PC default gateway wrong	ping gateway from PC
Intermittent adjacency drops	Hello/Dead timer mismatch	show ip ospf neighbor; show ip ospf interface

No adjacency on DS-1	ip routing not enabled	show running-config include ip routing
DS-1 Vlan10 SVI is down/down	No access port in VLAN 10 is up, or ports not assigned to VLAN 10	show vlan brief; show interfaces Vlan10
DS-1 SVI has no IP / IP overlap error	Tried to assign IPs to Gi0/0-Gi0/2 directly instead of using the Vlan10 SVI	show running-config interface Vlan10

7.4 Common Misconfiguration Checklist

- Wrong wildcard mask in network statement (e.g. using subnet mask instead of wildcard)
- OSPF process ID mismatch (local significance only, but confusing in logs)
- ip routing not configured on DS-1
- Interface in shutdown state
- Duplicate Router IDs (can cause intermittent adjacency issues)
- Area number mismatch between neighbors
- Authentication configured on one side only
- Assigning the same /24 IP address to multiple physical ports on DS-1 instead of using a single Vlan10 SVI
- Gi0/0-Gi0/2 on DS-1 not assigned to VLAN 10, leaving the Vlan10 SVI down

8. Review Questions

Answer the following questions to confirm your understanding:

1. What is the significance of the OSPF Process ID 123? Can two neighboring routers have different Process IDs and still form an adjacency?
2. R1 has a Loopback0 interface configured with 10.1.1.1/24. How does this affect OSPF router ID selection? What would R1's Router ID be without the loopback?
3. On the 192.168.1.0/24 broadcast segment, OSPF elects a DR and BDR. Which router becomes the DR by default and why? What happens if the DR fails?
4. What is the OSPF cost for R1 to reach PC4 (192.168.40.0/24) via this topology? Show your calculation.
5. If you configure 'passive-interface GigabitEthernet0/1' on R2, will 192.168.10.0/24 still be advertised into OSPF? Explain why.
6. What is the purpose of the 'write memory' command? What happens to your OSPF configuration if a router reboots without it?
7. Explain the difference between the OSPF LSDB and the routing table. Can a route be in the LSDB but not installed in the routing table?
8. DS-1 is a Layer 3 switch. What additional command is required before OSPF (or any routing) will function, and why?
9. Why must DS-1 use a VLAN SVI (Vlan10) instead of assigning IP addresses directly to Gi0/0, Gi0/1, and Gi0/2 for the 192.168.1.0/24 segment? What error occurs if you try the latter approach?

9. Lab Summary & Key Takeaways

Topic	Key Point
OSPF Process ID	Locally significant. Neighbors can differ. Process 123 used throughout this lab.
Router ID	R1 uses explicit 10.1.1.1 (loopback). Others auto-select highest active IP.
Network Statement	Uses wildcard mask. Matches interfaces — does not need to be an exact prefix.
Area 0	Single backbone area. All routers in this lab are internal routers.
DR/BDR	Elected on broadcast (Ethernet) segments. Reduces OSPF flooding overhead.
LSDB	Identical across all routers in an area after full convergence.
Cost Metric	Default cost = 10^8 / bandwidth. Lower is better. Can be manually overridden.
Passive Interface	Stops Hellos on stub segments. Network still advertised. Security best practice.
ip routing	Required on Layer 3 switches before any routing protocol will function.
VLAN SVI on DS-1	Used instead of routed ports so multiple links to the same subnet don't overlap. Single IP (192.168.1.254/24) on Vlan10 represents DS-1 on the OSPF backbone.
write memory	Always save config to NVRAM to survive reboots.

Lab Complete

You have successfully configured OSPFv2 Process 123 in Area 0 across R1, R2, R3, and DS-1. All four PC subnets are advertised and reachable. You have verified neighbor adjacencies, the LSDB, routing table entries, and end-to-end connectivity.